

Correlation Functions in a Finite-Size Kitaev Model

Chen-Chih Wang

Department of Physics, National Tsing Hua University, Hsinchu 30013, Taiwan
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Abstract

In this note, we discuss how to compute multi-spin correlation functions in a finite-size Kitaev model, which only contains four distinct plaquettes. These correlation functions are crucial in constructing the density matrix of the Kitaev ground state.

1 Foundations

1.1 Correlation Functions

As mentioned in our work [1], at the ground state, the multi-spin correlation function of the Kitaev model is finite if and only if it can be written in the form of a string. Therefore, we only need to discuss the following form of the correlation function

$$\left\langle \prod_{\langle ij \rangle \in \mathcal{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle$$

This kind of correlation function can be exactly evaluated in the Majorana representation. Substituting the transformation

$$\sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \rightarrow i b_i^{\alpha_{ij}} c_i i b_j^{\alpha_{ij}} c_j = \hat{u}_{ij} i c_j c_i$$

into the correlation function and taking the trivial gauge, we have

$$\left\langle \prod_{\langle ij \rangle \in \mathcal{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle = \left\langle \prod_{\langle ij \rangle \in \mathcal{D}} (-i) c_i c_j \right\rangle_M$$

1.2 Minimal Size for Non-trivial Topology

For numerical efficiency, we tend to choose the smallest possible system. However, the system must still be sufficiently large such that the topological order still exists. A requirement for this is that the ground state manifold should possess four independent topological sectors generated by global Wilson loops if the geometry of the system is a torus.

We start from the Wen plaquette model. There are two requirements for a Wen plaquette system to possess four topologically degenerate ground states:

- The lattice must contain an even number of rows or columns in both the horizontal and vertical directions.
- A global Wilson loop cannot be generated or destroyed by local plaquette operators.

The physical meaning of the first requirement is to preserve the checkerboard pattern of the lattice. If there are adjacent plaquettes (cells) that are of the same type (black or white, in a checkerboard sense), then the two types of plaquettes become identical, and the original two different Wilson loops become the same. This breakdown of the checkerboard pattern reduces the original fourfold topological ground state degeneracy down to twofold, which is not adequate to describe the physics of a toric code system.

According to the two requirements, **the smallest possible topologically non-trivial Wen plaquette model contains eight ($8 = 2 \times 4$) distinct plaquettes and eight sites**. A 2×2 checkerboard is not adequate for the system to have four topological sectors since it violates the second requirement¹.

¹The *global Wilson loop* is identical to a plaquette operator

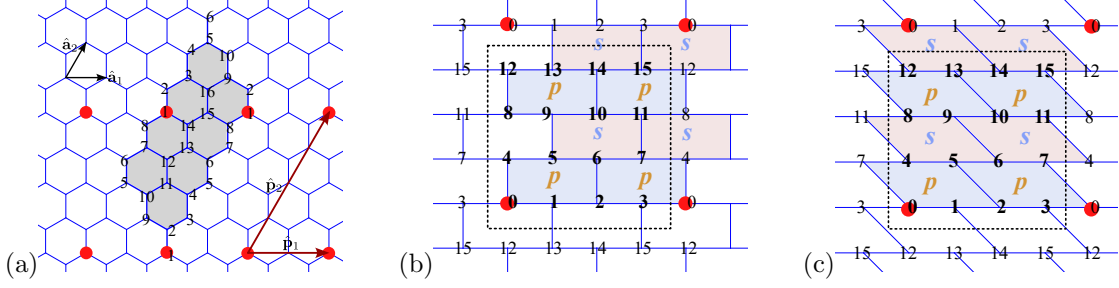


Figure 1: (a) The smallest topologically non-trivial system consists of 8 unit cells, which is equivalent to 16 spins. The two vectors that define the periodic boundary condition are $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, respectively. (b) For numerical simplicity, the honeycomb lattice is rearranged into a brick-wall geometry, and the spins are relabeled according to their Euclidean coordinates in the lattice. In both figures, red dots indicate identical lattice sites. (c) An equivalent choice.

2 Finite-Size Correlation

We only have to focus on the two-point correlation functions. Here we set that the system satisfies the **periodic boundary condition**, which is characterized by two independent vectors, $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, as illustrated in Fig.1.a. The geometry adopted here follows the one illustrated in Ref. [2]

2.1 Momentum States in Finite Systems

For slave fermions in the parton construction, the boundary condition of the system could be periodic (PBC) or anti-periodic (APBC), depending on which topological sector we choose [3]. Here, we only consider the PBC in both the horizontal and vertical directions.

$$f_{\sigma\mu\mathbf{x}} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_1)} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_2)} \quad (1)$$

To determine the allowed momentum states, we first Fourier transform the fermions as follows

$$f_{\sigma\mu\mathbf{x}} = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} f_{\sigma\mu\mathbf{k}}$$

With this expression, Eq. (1) implies that the allowed momentum $\mathbf{k}$ must satisfy

$$\begin{cases} \mathbf{k} \cdot \hat{\mathbf{p}}_1 = 2n_1\pi & ; \quad n_1 = 0, 1, \dots, N_h - 1 \\ \mathbf{k} \cdot \hat{\mathbf{p}}_2 = 2n_2\pi & ; \quad n_2 = 0, 1, \dots, N_v - 1 \end{cases} \quad (2)$$

In our case illustrated in Fig. 1, $N_h = 2$ and $N_v = 4$, representing the number of unit cells included in the system in the horizontal and vertical directions, respectively. Consider the unit vectors $\hat{\mathbf{a}}_1 = \hat{\mathbf{e}}_x$ and $\hat{\mathbf{a}}_2 = \frac{1}{2}\hat{\mathbf{e}}_x + \frac{\sqrt{3}}{2}\hat{\mathbf{e}}_y$. We can define the unit vectors in the momentum space as

$$\begin{cases} \hat{\mathbf{b}}_1 = 2\pi\hat{\mathbf{e}}_y \\ \hat{\mathbf{b}}_2 = \sqrt{3}\pi\hat{\mathbf{e}}_x - \pi\hat{\mathbf{e}}_y \end{cases}$$

such that $\hat{\mathbf{a}}_i \cdot \hat{\mathbf{b}}_j = 2\pi\delta_{ij}$. With these basis vectors, the allowed momentum states are

$$\mathbf{k} = \frac{n_1}{2}\hat{\mathbf{b}}_1 + \frac{n_2}{4}\hat{\mathbf{b}}_2 \quad ; \quad n_1 = 0, 1 \quad , \quad n_2 = 0, 1, 2, 3 \quad (3)$$

2.2 Two-Points Correlation Functions

The correlation functions can be calculated more easily in the normal fermion language. As defined in the appendix, the c -Majorana fermions can be written in terms of γ -fermions,

$$\begin{cases} c_{\mathbf{x}A} = \gamma_{\mathbf{x}}^\dagger + \gamma_{\mathbf{x}} = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} \left(\gamma_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} + \gamma_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} \right) \\ c_{\mathbf{x}B} = i(\gamma_{\mathbf{x}}^\dagger - \gamma_{\mathbf{x}}) = \frac{i}{\sqrt{N/2}} \sum_{\mathbf{k}} \left(\gamma_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} - \gamma_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} \right) \end{cases}$$

And the ground state is

$$|\psi\rangle = A \exp \left(\sum_{k \in \text{HBZ}} g_k \gamma_k^\dagger \gamma_{-k}^\dagger \right)$$

where

$$A = \prod_{k \in \text{HBZ}} \left(1 + |g_k|^2 \right)^{-1/2}$$

is the normalization constant. Similar to Nambu-Gor'kov Green's function, there are two types of fermion pairing, $AA(BB)$ and AB . Both types of correlation functions can be simplified to the problem of solving the γ -fermion correlation function as we will see later. The two γ -fermion correlation functions we have to know are $\langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle$ and $\langle \gamma_{\mathbf{k}}^\dagger \gamma_{k'} \rangle$. Their analytical forms are given as follows

$$\begin{cases} \langle \gamma_{\mathbf{k}}^\dagger \gamma_{k'} \rangle = \frac{|g_k|^2}{1 + |g_k|^2} \delta_{kk'} = v_k^2 \delta_{kk'} = \frac{1}{2} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \delta_{kk'} \\ \langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle = \frac{g_k}{1 + |g_k|^2} \delta_{k, -k'} = i \text{sgn}(I_k) u_k v_k \delta_{k, -k'} = i \frac{I_k}{2 |s_{\mathbf{k}}|} \delta_{k, -k'} \end{cases}$$

and the fermionic commutation relations can give other two types of correlation functions

$$\begin{cases} \langle \gamma_{\mathbf{k}} \gamma_{k'}^\dagger \rangle = \delta_{kk'} - \langle \gamma_{k'}^\dagger \gamma_{\mathbf{k}} \rangle = u_k^2 \delta_{kk'} = \frac{1}{2} \left(1 + \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \delta_{kk'} \\ \langle \gamma_{\mathbf{k}}^\dagger \gamma_{k'}^\dagger \rangle = \langle \gamma_{k'} \gamma_{\mathbf{k}} \rangle^* = -i \text{sgn}(I_{k'}) u_{k'} v_{k'} \delta_{k', -k} = i \text{sgn}(I_k) u_k v_k \delta_{k, -k'} = \langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle \end{cases}$$

With these relations, we can start to compute the c -Majorana correlation functions. For AA -pairings, the correlation function is,

$$\begin{aligned} \langle c_{\mathbf{x}A} c_{\mathbf{x}'A} \rangle &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(\langle \gamma_{\mathbf{k}} \gamma_{-\mathbf{k}} \rangle + \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{\mathbf{k}}^\dagger \rangle + \langle \gamma_{\mathbf{k}} \gamma_{\mathbf{k}}^\dagger \rangle + \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{-\mathbf{k}} \rangle \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \\ &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} \\ &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \cos(\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})) \in \mathbb{R} \end{aligned}$$

The imaginary part is an odd function in the momentum space, hence it vanishes after the integration. However, this implies that the Majorana correlation function is a real number, which is impossible since

$$\langle c_i c_j \rangle^* = \langle c_j c_i \rangle = -\langle c_i c_j \rangle$$

which must be an imaginary number unless $i = j$. So we conclude that

$$\langle c_{\mathbf{x}A} c_{\mathbf{x}'A} \rangle = \langle c_{\mathbf{x}B} c_{\mathbf{x}'B} \rangle = \delta_{\mathbf{x}\mathbf{x}'} \quad (4)$$

As for AB -pairings, a similar calculation gives

$$\begin{aligned} \langle c_{\mathbf{x}A} c_{\mathbf{x}'B} \rangle &= \frac{i}{N/2} \sum_{\mathbf{k}} \left(\langle -\gamma_{\mathbf{k}} \gamma_{-\mathbf{k}} \rangle + \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{\mathbf{k}}^\dagger \rangle + \langle \gamma_{\mathbf{k}} \gamma_{\mathbf{k}}^\dagger \rangle - \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{-\mathbf{k}} \rangle \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \\ &= \frac{i}{N/2} \sum_{\mathbf{k}} \left(\frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} - i \frac{I_k}{|s_{\mathbf{k}}|} \right) e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} = \frac{i}{N/2} \sum_{\mathbf{k}} \frac{s_{\mathbf{k}}^*}{|s_{\mathbf{k}}|} e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} \end{aligned}$$

This result is purely imaginary, and can be further simplified,

$$\langle -i c_{\mathbf{x}A} c_{\mathbf{x}'B} \rangle = \frac{1}{N/2} \sum_{\mathbf{k}} \cos[\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x}) - \theta_{\mathbf{k}}] = \frac{1}{N/2} \sum_{\mathbf{k}} \frac{1}{|s_{\mathbf{k}}|} [R_{\mathbf{k}} \cos(\mathbf{k} \cdot \delta \mathbf{x}) + I_{\mathbf{k}} \sin(\mathbf{k} \cdot \delta \mathbf{x})] \quad (5)$$

where $s_{\mathbf{k}} = |s_{\mathbf{k}}| e^{i\theta_{\mathbf{k}}}$. Notice that the above formulation is general and can be used directly in the case of finite systems.

2.3 Two-Points Correlation Functions in a Finite-Size System

In the following discussions, we consider the lattice geometry illustrated in Fig. 1.c. In this situation, there are three methods for labeling the position of lattice sites.

- The first method is to assign a number to each lattice site one by one. The label is denoted by n in this note.
- In the lattice geometry illustrated in Fig. 1.c, the lattice sites can be sorted into layers labeled by l , and the horizontal position of a lattice site in each layer is labeled by s , standing for *site*. For example, the position of site number 7 is $(l, s) = (1, 3)$, and the position of site number 12 is $(l, s) = (3, 0)$. The integer s can be further decomposed as

$$s = 2s_q + s_r$$

where $s_q \in \mathbb{N}$ and $s_r = 0 \vee 1$ is the remainder.

- With the Bravais lattice site expressed as $\mathbf{x} = x\hat{\mathbf{a}}_1 + y\hat{\mathbf{a}}_2$, each lattice site corresponds to a unique pair $(\mathbf{x}, \mu)$, where $\mu = 0, 1$, with 0 representing sublattice B and 1 representing A , respectively.
- The following equations describe the relationships among the three labeling methods listed above.

$$\begin{cases} n = s + lN_h \\ \mu = n \mod 2 \\ x = s_q + s_r \\ y = l - s_r \end{cases}$$

With this definition, the factor $\mathbf{k} \cdot \delta\mathbf{x}$ in Eq. (5) can be written as

$$\mathbf{k} \cdot \delta\mathbf{x} = \frac{n_1}{2}\delta x + \frac{n_4}{4}\delta y$$

where n_1 and n_2 are defined by the expression Eq. (3) of momentum states, and δx and δy are components of the position $\mathbf{x}$ represented in terms of lattice unit vectors $\mathbf{x} = \delta x\hat{\mathbf{a}}_1 + \delta y\hat{\mathbf{a}}_2$. Using this representation, the summation $\sum_{\mathbf{k}}$ in Eq. (5) can be written as $\sum_{n_1, n_2}$. The only difference is that the momentum is now represented in terms of integers n_1 and n_2 .

3 String Operators

3.1 Constructing String Operators

To define string operators, we have to define the **order** of multiplying the assembly of $\mathbb{Z}_2$ dimer operators to make each string operator uniquely defined.

Given the fact that a class $[\mathfrak{D}] \in \mathbf{G}$ in the **good set** $\mathbf{G}$, in which $\text{EP}(\mathfrak{D})$ contains the same amounts of the type- A and type- B endpoints,

$$\prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right)$$

Algorithm 1 Sorting lattice sites into A and B sublattices

```

function SUBLATID( $n$ )                                 $\triangleright$  Define a function that identifies in which sublattice  $\mu$  the site  $n \in \mathcal{N}$  is.
    return  $\mu$ 
end function
 $\text{EP}_A(\mathfrak{D}) \leftarrow \{\}; \text{EP}_B(\mathfrak{D}) \leftarrow \{\}$                  $\triangleright$  The sets that contains type- $A$  and type- $B$  sites, respectively
for  $n \in \mathfrak{D}$  do
    if SUBLATID( $n$ ) =  $A$  then
         $\text{EP}_A(\mathfrak{D}) \leftarrow \text{EP}_A(\mathfrak{D}) \cup \{n\}$ 
    else if SUBLATID( $n$ ) =  $B$  then
         $\text{EP}_B(\mathfrak{D}) \leftarrow \text{EP}_B(\mathfrak{D}) \cup \{n\}$ 
    end if
end for

```

Algorithm 2 Generate $\text{St}(\mathfrak{D})$

```
St( $\mathfrak{D}$ )  $\leftarrow \{\}$ 
for  $i = 0, i \leq |\text{EP}(\mathfrak{D})|/2, i++$  do
     $n_a \leftarrow \text{EP}_A(\mathfrak{D})[i]; n_b \leftarrow \text{EP}_B(\mathfrak{D})[i]$ 
    St( $\mathfrak{D}$ )  $\leftarrow \text{St}(\mathfrak{D}) \cup \{(n_a, n_b)\}$ 
end for
```

Algorithm 3 Dimer operators

Require: n and n' must be nearest neighbor, and $\mu(n) = A$ by default

```
( $l, s$ )  $\leftarrow \text{Dic}(n); (l', s') \leftarrow \text{Dic}(n')$ 
if  $l' = l - 1$  then
    dimer( $n, n'$ )  $= \sigma_n^z \sigma_{n'}^z$ 
else if  $s' = s + 1$  then
    dimer( $n, n'$ )  $= \sigma_n^x \sigma_{n'}^x$ 
else if  $s' = s - 1$  then
    dimer( $n, n'$ )  $= \sigma_n^y \sigma_{n'}^y$ 
end if
```

Algorithm 4 Generate the string operator of a single string

```
( $l_a, s_a$ )  $\leftarrow \text{Dic}(n_a); (l_b, s_b) \leftarrow \text{Dic}(n_b)$  ▷ The function Dic translates a lattice label  $n$  into the layer-site representation
 $\Sigma_{\mathfrak{D}} \leftarrow 1$  ▷ Initialize the string operator
( $l, s$ )  $\leftarrow (l_a, s_a)$  ▷ Initialize the moving point
while ( $l, s$ )  $\neq (l_b, s_b)$  do
    if  $l_b = l_a$  then
        if  $s_b > s_a$  then
            ( $l', s'$ )  $\leftarrow (l, s + 1)$ 
        else if  $s_b < s_a$  then
            ( $l', s'$ )  $\leftarrow (l, s - 1)$ 
        end if
    else if  $l_b > l_a$  then ▷ Move the moving point to the layer  $l_b$  first, and then shift it to the right site  $s_b$ .
        if the current moving point is in  $B$  then
            ( $l', s'$ )  $\leftarrow (l + 1, s - 1)$ 
        else if the current moving point is in  $A$  then
            ( $l', s'$ )  $\leftarrow (l, s + 1)$ 
        end if
    else if  $l_b < l_a$  then ▷ Move the moving point to the layer  $l_b$  first, and then shift it to the right site  $s_b$ .
        if the current moving point is in  $B$  then
            ( $l', s'$ )  $\leftarrow (l - 1, s + 1)$ 
        else if the current moving point is in  $A$  then
            ( $l', s'$ )  $\leftarrow (l, s + 1)$ 
        end if
    end if
     $n \leftarrow \text{Dic}^{-1}(l, s); n' \leftarrow \text{Dic}^{-1}(l', s')$ 
     $\Sigma_{\mathfrak{D}} \leftarrow \Sigma_{\mathfrak{D}} \cdot \text{dimer}(n, n')$ 
    ( $l, s$ )  $\leftarrow (l', s')$ 
end while
```

Notice that the periodic boundary condition requires

$$s + N_h = s, \quad l + N_v = l$$

Therefore, the function Dic^{-1} must satisfy

$$\text{Dic}^{-1}(s + N_h, l) = \text{Dic}^{-1}(s, l + N_v) = \text{Dic}^{-1}(s, l)$$

3.2 String Correlation Functions

$$\prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right)$$

We have required $\mathcal{C} \in \mathbf{G}$. That is to say, each string $\mathcal{C}$ connects a type- A and a type- B endpoint.

$$\begin{aligned} \prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} &\rightarrow \prod_{\langle ij \rangle \in \mathcal{C}} (-i) c_i c_j = (-i)^{|\mathcal{C}|} (-1)^{(|\mathcal{C}|-1)/2} c_{a_c} c_{b_c} \\ &= -i (-1)^{|\mathcal{C}|(|\mathcal{C}|-1)/2} (-1)^{(|\mathcal{C}|-1)/2} c_{a_c} c_{b_c} \\ &= -i (-1)^{(|\mathcal{C}|+1)^2/2} c_{a_c} c_{b_c} \end{aligned}$$

In the derivation, we have used the identity $i^n = (-i)^n = (-1)^{n(n-1)/2}$ if n is an even integer. The sign factor $(-1)^{(|\mathcal{C}|-1)/2}$ is due to reversing the direction of dimers to make them point from B to A . Notice that in each dimer $\sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}}$, the former index i represents the lattice site in the A -sublattice, while the latter one j stands for the B -sublattice, and the direction of the string $\mathcal{C}$ is always from its type- A endpoint to the type- B one. At the same time, because the two endpoints of each string are in different sublattices, considering the honeycomb lattice geometry, the length $|\mathcal{C}|$ of each string $\mathcal{C}$ must be an odd number. With this property, the factor $(|\mathcal{C}|+1)^2/2$ that appears in the exponent of the sign factor must be a multiple of 4, making the sign $(-1)^{(|\mathcal{C}|+1)^2/2}$ trivial. Therefore, we eventually obtain the correspondence between a string operator and its Majorana representation, which depends only on the endpoints a_c and b_c if $\mathcal{C}$ is a *good* string

$$\prod_{\langle ij \rangle \in \mathcal{C} \subset \mathbf{G}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \rightarrow -i c_{a_c} c_{b_c} \quad (6)$$

$$\prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right) \rightarrow \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} (-i c_{a_c} c_{b_c}) \quad (7)$$

$$\mathfrak{C}_{\mathfrak{D}} = \left\langle \prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \rightarrow \left\langle \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} (-i c_{a_c} c_{b_c}) \right\rangle_M = \begin{vmatrix} \langle -i c_{a_1} c_{b_1} \rangle_M & \langle -i c_{a_1} c_{b_2} \rangle_M & \langle -i c_{a_1} c_{b_3} \rangle_M & \cdots \\ \langle -i c_{a_2} c_{b_1} \rangle_M & \langle -i c_{a_2} c_{b_2} \rangle_M & \langle -i c_{a_2} c_{b_3} \rangle_M & \cdots \\ \langle -i c_{a_3} c_{b_1} \rangle_M & \langle -i c_{a_3} c_{b_2} \rangle_M & \langle -i c_{a_3} c_{b_3} \rangle_M & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} \quad (8)$$

4 Plaquette Operators

A Exactly Solving for the Ground State

According to equation (29) of Ref. [4], we can diagonalize the Kitaev Hamiltonian in the momentum space if the gauge is fixed to be $u_{ij} = +1 \forall (ij)$,

$$\hat{H} = \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger U(\mathbf{k}) \psi_{\mathbf{k}} = \frac{1}{2} \sum_{\mathbf{k}} \left[i s_{\mathbf{k}} d_{\mathbf{k},A}^\dagger d_{\mathbf{k},B} - i s_{\mathbf{k}}^* d_{\mathbf{k},B}^\dagger d_{\mathbf{k},A} \right]$$

where

$$\begin{aligned} \psi_{\mathbf{k}} &\equiv \frac{1}{\sqrt{2N}} \sum_{\mu} e^{-i\mathbf{k} \cdot \mathbf{r}_{\mu}} \begin{pmatrix} c_{\mu,A} \\ c_{\mu,B} \end{pmatrix} = \begin{pmatrix} d_{\mathbf{k},A} \\ d_{\mathbf{k},B} \end{pmatrix} \\ U(\mathbf{k}) &= \frac{1}{2} \begin{pmatrix} 0 & i s_{\mathbf{k}} \\ -i s_{\mathbf{k}}^* & 0 \end{pmatrix}, \quad s_{\mathbf{k}} = 2 \left(J_x e^{i\mathbf{k} \cdot (\mathbf{a}_2 - \mathbf{a}_1)} + J_y e^{i\mathbf{k} \cdot \mathbf{a}_2} + J_z \right) \end{aligned}$$

Note that the index μ above is the Bravais lattice point, and A, B represent the sublattice. If we further define a normal fermion operator,

$$\gamma_{\mu} \equiv \frac{1}{2} (c_{\mu,A} + i c_{\mu,B}) \quad (9)$$

then its Fourier transform would be

$$\gamma_k = \frac{1}{\sqrt{2N}} \sum_{\mu} \gamma_{\mu} e^{-ik \cdot r_{\mu}} = \frac{1}{2} (d_{k,A} + id_{k,B})$$

In this way, the operator d can be expressed in terms of γ fermion,

$$\begin{cases} d_{k,A} = \gamma_{-\mathbf{k}}^{\dagger} + \gamma_k \\ d_{k,B} = i(\gamma_{-\mathbf{k}}^{\dagger} - \gamma_k) \end{cases}$$

Using the γ operators to rewrite the Hamiltonian gives

$$\hat{H} = \sum_{\mathbf{k}} \left[R_{\mathbf{k}} \cdot (\gamma_{\mathbf{k}}^{\dagger} \gamma_{\mathbf{k}} - \gamma_{-\mathbf{k}} \gamma_{-\mathbf{k}}^{\dagger}) - i I_{\mathbf{k}} \cdot (\gamma_k^{\dagger} \gamma_{-\mathbf{k}}^{\dagger} - \gamma_{-\mathbf{k}} \gamma_k) \right] = \sum_{\mathbf{k}} \phi_{\mathbf{k}}^{\dagger} \mathbf{H}(k) \phi_{\mathbf{k}}$$

where $R_{\mathbf{k}} \equiv \text{Re} \{s_{\mathbf{k}}\}$ and $I_k \equiv \text{Im} \{s_{\mathbf{k}}\}$, and

$$\mathbf{H}(k) \equiv \begin{pmatrix} R_{\mathbf{k}} & -i I_{\mathbf{k}} \\ i I_k & -R_{\mathbf{k}} \end{pmatrix}, \quad \phi_k \equiv \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma_{-\mathbf{k}}^{\dagger} \end{pmatrix}$$

Since

$$\mathbf{H}(k)^2 = (R_{\mathbf{k}}^2 + I_k^2) \mathbb{1} = |s_{\mathbf{k}}|^2 \mathbb{1}$$

we can see that the eigenvalues are $\epsilon_k = \pm |s_{\mathbf{k}}|$, which is consistent with Kitaev's result (equation (32) in Ref.[?]). To diagonalize the matrix $\mathbf{H}$ we have to apply the Bogoliubov transformation,

$$U = \frac{1}{\sqrt{2|s_{\mathbf{k}}|}} \begin{pmatrix} \text{sgn}(I_k) \sqrt{|s_{\mathbf{k}}| + R_{\mathbf{k}}} & -i \sqrt{|s_{\mathbf{k}}| - R_{\mathbf{k}}} \\ i \sqrt{|s_{\mathbf{k}}| - R_{\mathbf{k}}} & -\text{sgn}(I_k) \sqrt{|s_{\mathbf{k}}| + R_{\mathbf{k}}} \end{pmatrix} = \begin{pmatrix} \text{sgn}(I_k) u_k & -i v_k \\ i v_k & -\text{sgn}(I_k) u_k \end{pmatrix} = v_k \sigma^y + \text{sgn}(I_k) u_k \sigma^z$$

such that²

$$U \mathbf{H} U^{\dagger} = \begin{pmatrix} |s_{\mathbf{k}}| & 0 \\ 0 & -|s_{\mathbf{k}}| \end{pmatrix}$$

Therefore, the excitation that diagonalize the Hamiltonian is

$$\begin{pmatrix} a_k \\ a_{-k}^{\dagger} \end{pmatrix} = U \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma_{-\mathbf{k}}^{\dagger} \end{pmatrix} = \begin{pmatrix} \text{sgn}(I_k) u_k \gamma_{\mathbf{k}} - i v_k \gamma_{-\mathbf{k}}^{\dagger} \\ i v_k \gamma_{\mathbf{k}} - \text{sgn}(I_k) u_k \gamma_{-\mathbf{k}}^{\dagger} \end{pmatrix} \quad (10)$$

As a result, the Kitaev ground state is the vacuum state of the Bogoliubov fermion, i.e., $a_k |M\rangle = 0$, which means $|M\rangle$ can be written as a BCS ground state just as we met in the BCS theory. The ground state can be written as the coherent state³ of the γ -fermion pair,

$$\begin{aligned} |M\rangle &= A[g] \exp \left(\sum_{\mathbf{k}} g_{\mathbf{k}} \gamma_{\mathbf{k}}^{\dagger} \gamma_{-\mathbf{k}}^{\dagger} \right) |\text{vac}_{\gamma}\rangle \\ &= A[g] \exp \left(\sum_{\mu\nu} g_{\mu\nu} \gamma_{\mu}^{\dagger} \gamma_{\nu}^{\dagger} \right) |\text{vac}_{\gamma}\rangle \\ &= A[g] \left[\mathbb{1} + \sum_{\mu\nu} g_{\mu\nu} \gamma_{\mu}^{\dagger} \gamma_{\nu}^{\dagger} + \frac{1}{2!} \sum_{\mu\nu\kappa\lambda} g_{\mu\nu} g_{\kappa\lambda} \gamma_{\mu}^{\dagger} \gamma_{\nu}^{\dagger} \gamma_{\kappa}^{\dagger} \gamma_{\lambda}^{\dagger} + \dots \right] |\text{vac}_{\gamma}\rangle \end{aligned}$$

where $|\text{vac}_{\gamma}\rangle$ is the vacuum state of the γ -fermions, and $g_{\mu\nu} = \frac{1}{2N} \sum_{\mathbf{k}} g_{\mathbf{k}} e^{-ik \cdot (r_{\mu} - r_{\nu})}$.

²Those parameters in the matrix U have the following properties: (i) $u_k = u_{-k}$, (ii) $v_k = v_{-k}$, and (iii) $\text{sgn}(I_k) = -\text{sgn}(I_{-k})$

³The coefficient g_k can be represented in terms of u_k and v_k . As I have mentioned, the BCS-like ground state is expected to be the vacuum state of a_k fermions for all possible k , thus, we have

$$(\text{sgn}(I_k) u_k \gamma_{\mathbf{k}} - i v_k \gamma_{-\mathbf{k}}^{\dagger}) \prod_k (\mathbb{1} + g_{\mathbf{k}} \gamma_{\mathbf{k}}^{\dagger} \gamma_{-\mathbf{k}}^{\dagger}) |\text{vac}_{\gamma}\rangle = 0$$

Based on this, we obtain that

$$\text{sgn}(I_k) u_k g_k - i v_k = 0$$

therefore

$$g_k = \text{sgn}(I_k) i \frac{v_k}{u_k} = \text{sgn}(I_k) i \sqrt{\frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{|s_{\mathbf{k}}| + R_{\mathbf{k}}}} = i \frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{I_k} \quad (11)$$

B Sum Rules for Correlation Functions

$$\rho = \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}$$

$$\mathcal{P}^{(p)} = \left(\frac{\mathbb{1} + \tilde{W}_v}{2} \right) \left(\frac{\mathbb{1} + \tilde{W}_h}{2} \right) \prod_p \left(\frac{\mathbb{1} + W_p}{2} \right)$$

$$\begin{aligned} \rho^2 &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\ &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \frac{1}{2} [\varphi(\mathfrak{D}, \mathfrak{D}') + \varphi(\mathfrak{D}', \mathfrak{D})] \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\ &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\ &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}^c]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}^c \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \vartheta(\mathfrak{D}^c \ominus \mathfrak{D}', \mathfrak{D}') \Sigma_{\mathfrak{D}^c} \\ &= \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \left(\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \eta_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}') \right) \Sigma_{\mathfrak{D}} \\ &\quad \frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \vartheta(\mathfrak{D}, \mathfrak{D}') \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} = \mathfrak{C}_{\mathfrak{D}} \end{aligned} \tag{12}$$

$$\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}}^2 = 1 \tag{13}$$

notice that $\eta_{\mathfrak{D}} = 1$ if $[\mathfrak{D}] \in \mathbf{G}$

$$\vartheta(\mathfrak{D}, \mathfrak{D}') = \frac{1}{2} [1 + \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D} \ominus \mathfrak{D}'}] \varphi(\mathfrak{D}, \mathfrak{D}') = \vartheta(\mathfrak{D}', \mathfrak{D}) = 0 \text{ or } \pm 1$$

$$\varphi(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \eta_{\mathfrak{D}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \varphi(\mathfrak{D}', \mathfrak{D}) = \eta_{\mathfrak{D}'} \varphi(\mathfrak{D}, \mathfrak{D}')$$

$$\vartheta(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \frac{1}{2} [1 + \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D}}] \varphi(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \eta_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}')$$

Lemma B.1. If $[\mathfrak{D}] \in \mathbf{G}$, then the following identity holds

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} = \delta_{[\mathfrak{D}], [\emptyset]}$$

Proof. By definition, the factor $\eta_{\mathfrak{D}}$ can be expressed as $\eta_{\mathfrak{D}} = (-1)^{\Delta(\mathfrak{D})/2}$, where $\Delta(\mathfrak{D}) \equiv n_a(\mathfrak{D}) - n_b(\mathfrak{D})$.

$$\Delta(\mathfrak{D} \ominus \mathfrak{D}') = \Delta(\mathfrak{D}) + \Delta(\mathfrak{D}') - n_a(\mathfrak{D} \cap \mathfrak{D}') + n_b(\mathfrak{D} \cap \mathfrak{D}')$$

$$\eta_{\mathfrak{D} \ominus \mathfrak{D}'} = (-1)^{\Delta(\mathfrak{D} \ominus \mathfrak{D}')} = \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} (-1)^{n_a(\mathfrak{D} \cap \mathfrak{D}')} (-1)^{n_b(\mathfrak{D} \cap \mathfrak{D}')} = (-1)^{n_a(\mathfrak{D} \cap \mathfrak{D}')} (-1)^{n_b(\mathfrak{D} \cap \mathfrak{D}')}$$

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} = \sum_{[\mathfrak{D}'] \in \mathbf{G}} (-1)^{n_a(\mathfrak{D} \cap \mathfrak{D}')} (-1)^{n_b(\mathfrak{D} \cap \mathfrak{D}')}$$

■

Lemma B.2. If $[\mathfrak{D}] \in \mathbf{G}$, then the following identity holds

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \Sigma_{\mathfrak{D}'} \Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \delta_{[\mathfrak{D}], [\emptyset]} \mathbb{1}$$

Proof. According to the identity $\varphi(\mathfrak{D}, \mathfrak{D}') = \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \varphi(\mathfrak{D}', \mathfrak{D})$, we have the commutation relation

$$\Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \Sigma_{\mathfrak{D}'} \Sigma_{\mathfrak{D}}$$

In the above identity, we have already considered $[\mathfrak{D}], [\mathfrak{D}'] \in \mathbf{G}$ in our case. With this relation, we could rewrite the summation in the equation we wanted to prove

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \Sigma_{\mathfrak{D}'} \Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \sum_{[\mathfrak{D}'] \in \mathbf{G}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \Sigma_{\mathfrak{D}}$$

■

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